



ASCHAM SCHOOL

YEAR 12

MATHEMATICS EXTENSION 1

TRIAL EXAMINATION 2015

GENERAL INSTRUCTIONS

5 minutes reading time.

Working time: 2 hours.

Use black or blue pen.

A table of standard integrals is provided separately.

Approved calculators and templates may be used.

Total Marks - 70

Section A – Multiple Choice (1 mark each)

Attempt Questions 1 to 10.

Allow approximately 15 minutes.

Select answers on the separate multiple choice sheet provided.

Write your BOS number on the multiple choice sheet.

Section B – Questions 11 – 14 (15 marks each)

Allow 1 hour 45 minutes.

Start each question in a new booklet.

If you use a second booklet for a question, place it inside the first.

Label extra booklets for the same question as, for example, Q11-2 etc.

Write your BOS number and question number on each booklet.

Blank page

Section A - Multiple choice (10 marks)

(Mark the correct answer on the sheet provided.)

1. The interval AB between $A(2, -1)$ and $B(-6, 3)$ is divided internally by the point P in the ratio $1 : 3$. The correct coordinate of P is given by:

A) $\left(\frac{1 \times 2 + 3 \times -6}{1 + 3}, \frac{1 \times -1 + 3 \times 3}{1 + 3}\right)$ B) $\left(\frac{1 \times 2 - 3 \times -6}{1 - 3}, \frac{1 \times -1 - 3 \times 3}{1 - 3}\right)$

C) $\left(\frac{1 \times -6 + 3 \times 2}{1 + 3}, \frac{1 \times 3 + 3 \times -1}{1 + 3}\right)$ D) $\left(\frac{1 \times -6 - 3 \times 2}{1 - 3}, \frac{1 \times 3 - 3 \times -1}{1 - 3}\right)$

2. Find $\int \frac{1}{\sqrt{16 - 9x^2}} dx$.

A) $\sin^{-1}\left(\frac{x}{4}\right) + C$ B) $\sin^{-1}\left(\frac{3x}{4}\right) + C$

C) $\frac{1}{3} \sin^{-1}\left(\frac{3x}{4}\right) + C$ D) $\frac{1}{4} \sin^{-1}\left(\frac{3x}{4}\right) + C$

3. What is the natural domain of $f(x) = \log_e(\cos^{-1} x)$?

A) $0 < x \leq 1$ B) $-1 < x \leq 1$

C) $0 \leq x < 1$ D) $-1 \leq x < 1$

4. Which of the following is a valid solution to the differential equation

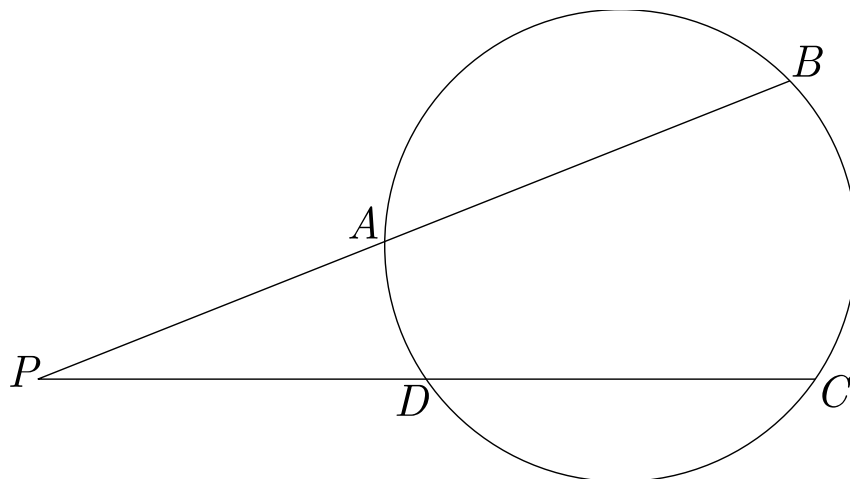
$\frac{dP}{dt} = -k(P - M)$, where k and M are constants?

A) $P = Me^{kt}$ B) $P = Me^{-kt}$

C) $P = M + e^{kt}$ D) $P = M + e^{-kt}$

(Section A continues on the next page...)

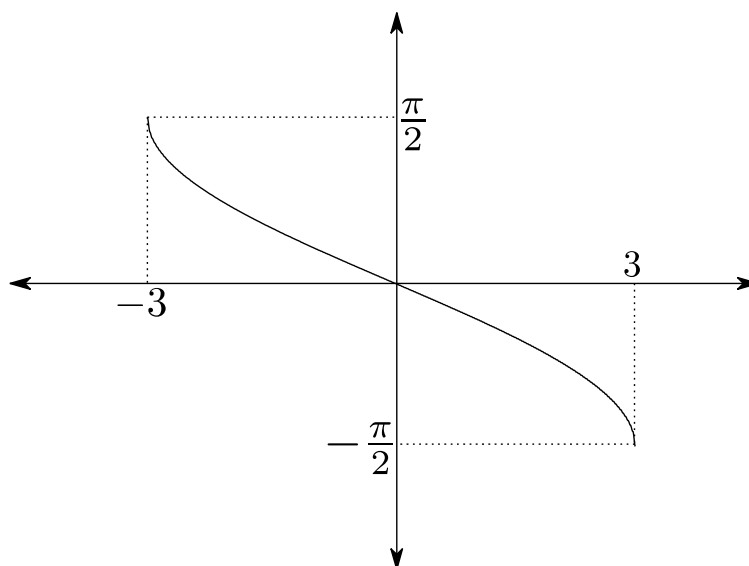
5.



Which is the correct relation for the intervals in the diagram above?

- A) $PA \times AB = PD \times DC$ B) $PA \times DC = PD \times AB$
 C) $PA \times PB = PD \times PC$ D) $PA \times PC = PD \times PB$

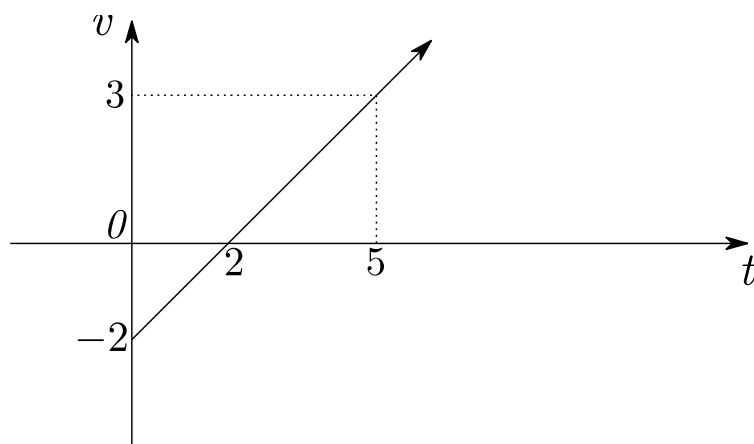
6. Which of the following is the correct function for the graph below?



- A) $y = -\sin^{-1}\left(\frac{x}{3}\right)$ B) $y = -\cos^{-1}\left(\frac{x}{3}\right)$
 C) $y = -\sin^{-1}(3x)$ D) $y = -\cos^{-1}(3x)$

(Section A continues on the next page...)

9. A particle moving on a straight line is depicted by the following velocity-time graph. What distance does it travel in the first 5 seconds of motion?



- A) 2.5 units B) 3 units
- C) 5 units D) 6.5 units

(Section A continues on the next page...)

10. Given that α , β and γ are the roots of $x^3 - 3x^2 - 5x + 6 = 0$, what is the value of $\alpha^2 + \beta^2 + \gamma^2$?

A) -1

B) 4

C) 14

D) 19

(End of Section A. Question 11 begins on the next page.)

Section B (60 marks)

Question 11 (Begin and label a new booklet.)

(15 marks)

- a) Find the acute angle between the lines $x - 2y + 1 = 0$ and $y = 5x - 4$. Give your answer in radians correct to two decimal places. [2]

- b) Solve the inequality $\frac{3x+2}{x-2} > 1$. [3]

- c) Use Newton's method to find a better approximation to the root of $\log_e x - \sin x = 0$, given that the root is near $x = 2$. (x is a radian)
Give your answer to 2 decimal places. [3]

- d) Let $P(x) = (x+1)(x-3)Q(x) + a(x+1) + b$, where $Q(x)$ is a polynomial and a and b are real numbers.

When $P(x)$ is divided by $(x+1)$, the remainder is -11 .

When $P(x)$ is divided by $(x-3)$, the remainder is 1 .

- i) What is the value of b ? [1]

- ii) What is the remainder when $P(x)$ is divided by $(x+1)(x-3)$? [2]

- e) Prove by mathematical induction that [4]

$$\frac{1}{3} + \frac{1}{15} + \frac{1}{35} + \dots + \frac{1}{4n^2 - 1} = \frac{n}{2n + 1}$$

for positive integers n .

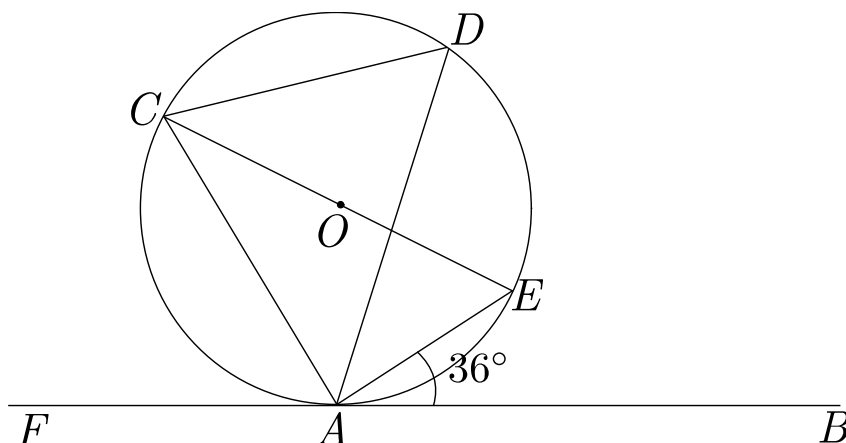
(End of Question 11.)

Question 12 (Begin and label a new booklet.)

(15 marks)

a) Evaluate $\int_0^{\frac{\pi}{8}} \cos^2 x \, dx$ [2]

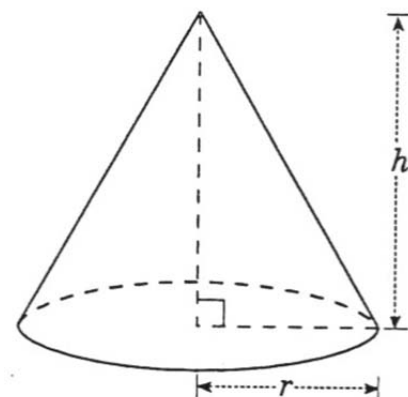
- b) FB is a tangent touching a circle at A . CE is the diameter, O is the centre and D lies on the circumference. $\angle BAE = 36^\circ$.



- i) Find the size of $\angle ACE$, giving reasons. [1]
- ii) Find the size of $\angle ADC$, giving reasons. [2]
- c) i) Show that the equation of tangent at point $P(2ap, ap^2)$ on the parabola $x^2 = 4ay$ is given by $px - y - ap^2 = 0$. [2]
- ii) S is the focus of the parabola and T is the point of intersection of the tangent and the y-axis.
- Prove that $SP = ST$. [2]
- iii) Hence show that $\angle SPT$ is equal to the acute angle between the tangent and the line through P parallel to the axis of the parabola. [2]

(Question 12 continues on the next page...)

d) Sand is poured at a rate of 2 cubic metres per minute. It forms a conical pile, with the angle at the apex of the cone equal to 60° . The height of the pile is h metres, and the radius of the base is r metres.



i) Show that $r = \frac{h}{\sqrt{3}}$. [1]

ii) Show that V , the volume of the pile, is given by $V = \frac{\pi h^3}{9}$. [1]

iii) Hence find the rate at which the height of the pile is increasing when the height of the pile is 3 metres. [2]

(End of Question 12.)

Question 13 (Begin and label a new booklet.)

(15 marks)

a) i) Find $\frac{d}{dx}(x \tan^{-1} x)$ [1]

ii) Hence evaluate $\int_0^1 \tan^{-1} x \, dx$ [2]

b) A particle is moving on a straight line with its displacement described by $x = \cos 2t + \sin 2t$, where t is in seconds.

i) Show that the particle is in simple harmonic motion, namely its acceleration has the form $\ddot{x} = -n^2 x$. [2]

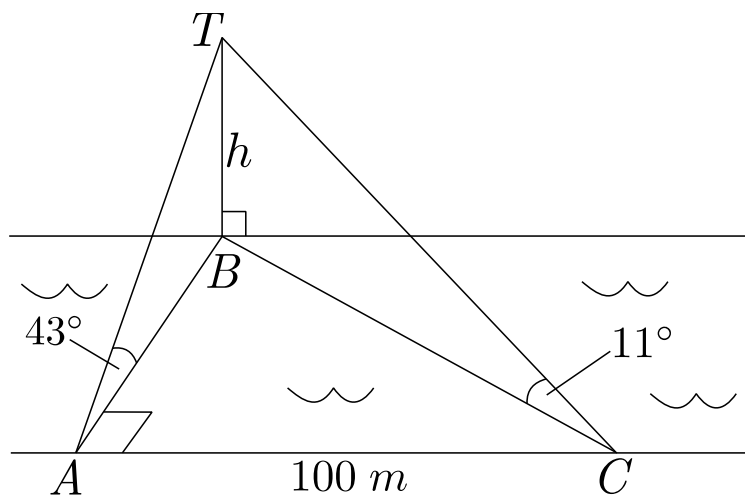
ii) State the period of motion. [1]

iii) By expressing $\cos 2t + \sin 2t$ in the form $R \cos(2t - \alpha)$, find the amplitude of the motion. [2]

iv) Hence or otherwise find the value of t at the first moment that the particle is stationary. [2]

(Question 13 continues on the next page...)

- c) A tree BT is observed directly across a creek with an angle of elevation of 43° from the ground. After walking 100 metres along the bank from A to C , the same tree now has an angle of elevation of 11° .



- i) Show that the creek has a width of $\frac{h}{\tan 43^\circ}$. [1]
- ii) Find the height of the tree to 2 decimal places. [4]

(End of Question 13.)

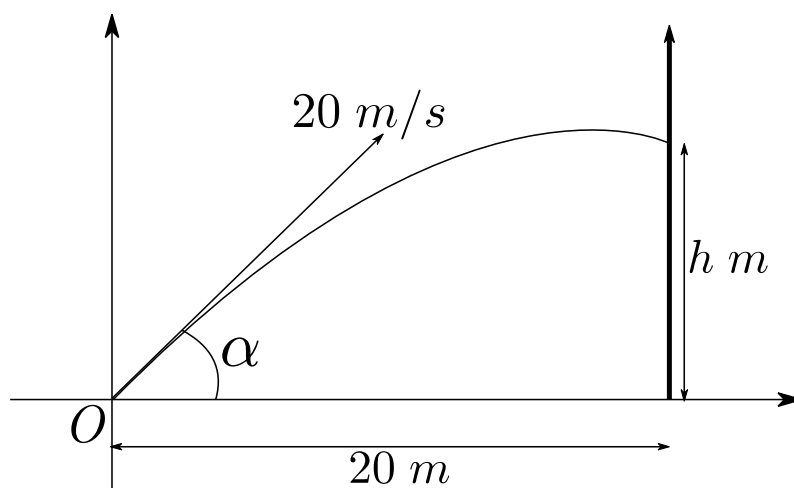
Question 14 (Begin and label a new booklet.)**(15 marks)**

a) By using the substitution $u = x^2 - 5$, or otherwise, find $\int \frac{x^3}{\sqrt{x^2 - 5}} dx$. [3]

b) Show that $\frac{1 - \tan x}{1 + \tan x} = \frac{1 - 2 \sin x \cos x}{\cos^2 x - \sin^2 x}$ [3]

c) A cricketer hits the ball from ground level with a speed of 20 m/s and an angle of elevation of α . It flies towards a high wall 20 metres away.

Take $g = 10 \text{ m/s}^2$



i) Given that the horizontal and vertical displacements at time t are, respectively (do not need to derive these):

$$x = 20t \cos \alpha$$

$$y = -5t^2 + 20t \sin \alpha$$

Show that the value of h , the height up the wall at which the ball will collide, is given by $h = -5 \sec^2 \alpha + 20 \tan \alpha$. [2]

ii) Show that the maximum value of h is obtained when $\tan \alpha = 2$. [2]

iii) Assuming $\tan \alpha = 2$, find the speed at which the ball will hit the wall. [2]

(Question 14 continues on the next page...)

d) i) State any discontinuities, if any exist, of the function $f(x) = \frac{1+e^x}{1-e^x}$. [1]

ii) It is given that $f'(x) = \frac{2e^x}{(1-e^x)^2}$.

Sketch $y = f(x)$, showing all important features. [2]

End of exam.

2015
Ext 1 Trial Solutions

Section A. Q1.C Q2.C Q3.D Q4.D Q5.C
Q6.A Q7.B Q8.C Q9.D Q10.D

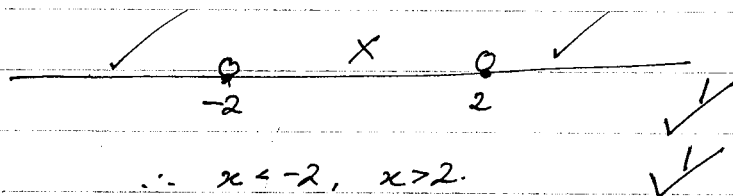
Q11. a) $x - 2y + 1 = 0$ $y = 5x - 4$ $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$
 $2y = x + 1$ $m_2 = 5$ $= \left| \frac{\frac{1}{2} - 5}{1 + \frac{5}{2}} \right|$ ✓
 $y = \frac{x}{2} + \frac{1}{2}$ $= \frac{9}{7}$
 $m_1 = \frac{1}{2}$ $\theta = 52.125^\circ$
 $\approx 0.91^\circ$ to 2d.p. ✓

b). $\frac{3x+2}{x-2} > 1$

Disc: $x = 2$.

Solutions: $3x+2 = x-2$

$2x = -4$
 $x = -2$ ✓



c) $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$ ✓

$f(2) \approx -0.21615 \dots$

$x_2 = 2 - \frac{-0.21615}{0.91614}$

$f'(x) = \frac{1}{x} - \cos x$

$= 2.2359 \dots$

$f'(2) = 0.91614 \dots$ ✓

≈ 2.24 to 2d.p. ✓

d) i) $P(-1) = -11$ by remainder theorem.

$\therefore -11 = 0 + 0 + b$

$b = -11$ ✓

ii) $P(3) = 1$

$P(3) = 0 + 4a + b$

$1 = 4a + b$ ✓

$1 = 4a - 11$

$4a = 12$ $a = 3$

Remainder is

$3(x+1) - 11 = 3x - 8$ ✓

Q11e) let $n=1$, $\frac{1}{3} = \frac{1}{2 \times 1 + 1}$

$\frac{1}{3} = \frac{1}{4 \times 1^2 - 1}$ true for $n=1$ ✓

Assume for $n=k$, $\frac{1}{3} + \frac{1}{15} + \dots + \frac{1}{4k^2 - 1} = \frac{k}{2k+1}$

RTP for $n=k+1$ $\frac{1}{3} + \frac{1}{15} + \dots + \frac{1}{4k^2 - 1} + \frac{1}{4(k+1)^2 - 1} = \frac{k+1}{2(k+1)+1}$ ~~XXXXXX~~

LHS = $\frac{k}{2k+1} + \frac{1}{4(k+1)^2 - 1}$ by assumption ✓

= $\frac{k}{2k+1} + \frac{1}{(2k+2-1)(2k+2+1)}$

= $\frac{k}{2k+1} + \frac{1}{(2k+1)(2k+3)}$

= $\frac{k(2k+3)+1}{(2k+1)(2k+3)}$

= $\frac{2k^2+3k+1}{(2k+1)(2k+3)}$

= $\frac{(2k+1)(k+1)}{(2k+1)(2k+3)}$

= $\frac{k+1}{2k+3}$

RHS = $\frac{k+1}{2(k+1)+1}$

= $\frac{k+1}{2k+3}$

✓✓

∴ LHS = RHS

∴ Statement true for $n=1, 2, \dots$ all positive integers n
by mathematical induction.

Q12. a) $\int_0^{\frac{\pi}{8}} \cos^2 x dx$

$= \frac{1}{2} \int_0^{\frac{\pi}{8}} 1 + \cos 2x dx$ ✓

$= \frac{1}{2} \left[x + \frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{8}}$

$= \frac{1}{2} \left(\frac{\pi}{8} + \frac{1}{2} \cdot \frac{1}{\sqrt{2}} \right) - \frac{1}{2}(0)$

$= \frac{\pi}{16} + \frac{1}{2\sqrt{2}}$ ✓

b) i) $\angle ACE = \angle EAB$ (angle in alternate segment)
 $= 36^\circ$

ii) $\angle CAE = 90^\circ$ (angle in semicircle)

$\therefore \angle CEA = 180 - 36 - 90$ (angle sum of triangle) ✓
 $= 54^\circ$

$\angle CDA = \angle CEA$ (angles standing on same arc are equal) ✓
 $= 54^\circ$

c) i) $\frac{dy}{dx} = \frac{dy/dp}{dx/dp}$

$= \frac{2ap}{2a}$

$= p$ ✓

Eqn tangent: $y - ap^2 = p(x - 2ap)$

$y - ap^2 = px - 2ap^2$

$y = px - ap^2$

$px - y - ap^2 = 0$ ✓

ii) $S: (0, a)$

$T: x=0, -y - ap^2 = 0 \Rightarrow y = -ap^2$

$\therefore T(0, -ap^2)$

$SP = \sqrt{(2ap)^2 + (ap^2 - a)^2}$

$= \sqrt{4a^2p^2 + a^2p^4 - 2a^2p^2 + a^2}$

$= \sqrt{a^2(p^4 + 2p^2 + 1)}$

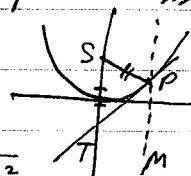
$= \sqrt{a^2(p^2 + 1)^2}$

$SP = a(p^2 + 1)$ ✓

$ST = |a - (-ap^2)|$

$= ap^2 + a$ ✓

$\therefore SP = ST$ ✓



iii) $\angle SPT = \angle STP$

(base $\angle$ s of isosceles Δ are equal) ✓

$\angle TPM = \angle STP$ (alternate $\angle$ s equal, $ST \parallel PM$) ✓

$\therefore \angle SPT = \angle TPM$ ✓
as required.

Q12d) i) $\tan 30 = \frac{r}{h}$
 $r = \frac{h}{\sqrt{3}}$ ✓

ii) $V = \frac{1}{3}\pi r^2 h$
 $= \frac{1}{3}\pi \times \left(\frac{h^2}{3}\right) \times h$
 $= \frac{\pi h^3}{9}$ ✓

iii) $\frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt}$
 $\frac{dV}{dt} = 2,$
 at $h=3,$ $\frac{dV}{dh} = \frac{\pi \times 3^2}{3}$
 $= 3\pi$

$V = \frac{\pi h^3}{9}$
 $\frac{dV}{dh} = \frac{3\pi h^2}{9}$
 $= \frac{\pi h^2}{3}$ ✓

$2 = 3\pi \times \frac{dh}{dt}$
 $\frac{dh}{dt} = \frac{2}{3\pi} \text{ m/min.}$ ✓

Q13 a) i) $\frac{d}{dx}(x \tan^{-1}x)$

$$= \tan^{-1}x + \frac{x}{1+x^2}$$

ii) $\therefore \int \tan^{-1}x + \frac{x}{1+x^2} dx = x \tan^{-1}x + C$

$$\int \tan^{-1}x dx + \int \frac{x}{1+x^2} dx = x \tan^{-1}x + C$$

$$\int \tan^{-1}x dx = x \tan^{-1}x - \int \frac{x}{1+x^2} dx$$

$$\int \tan^{-1}x dx = x \tan^{-1}x - \frac{1}{2} \log_e(1+x^2) + C$$

$$\int_0^1 \tan^{-1}x = \left[x \tan^{-1}x - \frac{1}{2} \log_e(1+x^2) \right]_0^1$$

$$= (1 \tan^{-1}1 - \frac{1}{2} \log_e(2)) - (0 - \log_e 1)$$

$$= \frac{\pi}{4} - \frac{1}{2} \log_e 2$$

b) i) $x = \cos 2t + \sin 2t$

$$\dot{x} = -2 \sin 2t + 2 \cos 2t$$

$$\ddot{x} = -4 \cos 2t - 4 \sin 2t$$

$$= -4(\cos 2t + \sin 2t)$$

$$= -4x$$

$$\therefore \text{SHM.}$$

ii) $\ddot{x} = -4x$, $n=2$, $T = \frac{2\pi}{2} = \pi$ sec.

iii) $\cos 2t + \sin 2t = R \cos(2t - \alpha)$

$$\text{LHS} = \sqrt{2} \left(\frac{1}{\sqrt{2}} \cos 2t + \frac{1}{\sqrt{2}} \sin 2t \right)$$

$$= \sqrt{2} (\cos 2t \cos \frac{\pi}{4} + \sin 2t \sin \frac{\pi}{4})$$

$$= \sqrt{2} \cos(2t - \frac{\pi}{4})$$

Amplitude: $\sqrt{2}$ units

iv) $\ddot{x} = -2\sqrt{2} \sin(2t - \frac{\pi}{4})$

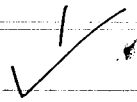
$$\ddot{x} = 0: \sin(2t - \frac{\pi}{4}) = 0$$

$$2t - \frac{\pi}{4} = 0, \pi, \dots$$

$$2t = \frac{\pi}{4}, \frac{5\pi}{4}, \dots \quad t = \frac{\pi}{8} \text{ is the first time } \ddot{x} = 0.$$

$$c) i) \frac{h}{BA} = \tan 43$$

$$\therefore BA = \frac{h}{\tan 43^\circ}$$



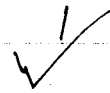
ii) By Pythagoras:

$$BC = \frac{h}{\tan 11^\circ}$$



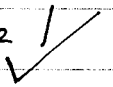
$$AC^2 + AB^2 = BC^2$$

$$\therefore 10000 + \frac{h^2}{\tan^2 43} = \frac{h^2}{\tan^2 11}$$



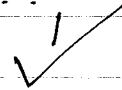
$$\therefore 10000 = h^2 \left(\frac{1}{\tan^2 11} - \frac{1}{\tan^2 43} \right)$$

$$10000 = \cancel{0.445382} h^2 \quad 25.31646 \dots h^2$$



$$h^2 = 394.9999 \dots$$

$$h \doteq 19.87 \text{ m.}$$



$$\textcircled{Q14a)} \int \frac{x^3}{\sqrt{x^2-5}} dx \quad u=x^2-5, \quad x^2=u+5 \\ du=2x dx$$

$$= \frac{1}{2} \int \frac{2x \cdot x^2}{\sqrt{x^2-5}} dx$$

$$= \frac{1}{2} \int \frac{u+5}{\sqrt{u}} du$$

$$= \frac{1}{2} \int u^{\frac{1}{2}} + 5u^{-\frac{1}{2}} du$$

$$= \frac{1}{2} \frac{u^{\frac{3}{2}}}{\frac{3}{2}} + \frac{5}{2} \frac{u^{\frac{1}{2}}}{\frac{1}{2}} + C$$

$$= \frac{2}{3} \frac{1}{2} u^{\frac{3}{2}} + \frac{5}{2} \cdot \frac{2}{1} u^{\frac{1}{2}} + C$$

$$= \frac{1}{3} u^{\frac{3}{2}} + 5u^{\frac{1}{2}} + C \\ = \frac{1}{3} (x^2-5)^{\frac{3}{2}} + 5(x^2-5)^{\frac{1}{2}} + C$$

$$b) \text{ LHS} = \frac{1-2\sin x \cos x}{\cos^2 x - \sin^2 x}$$

$$\text{RHS} = \frac{1-\tan x}{1+\tan x}$$

$$\text{LHS} = \frac{\left(1 - \frac{\sin x}{\cos x}\right) \times \cos x}{\left(1 + \frac{\sin x}{\cos x}\right) \times \cos x}$$

$$= \frac{\cos x - \sin x}{\cos x + \sin x}$$

$$= \frac{(\cos x - \sin x)^2}{\cos^2 x - \sin^2 x}$$

$$= \frac{\cos^2 x + \sin^2 x - 2\sin x \cos x}{\cos^2 x - \sin^2 x}$$

$$= \frac{1 - 2\sin x \cos x}{\cos^2 x - \sin^2 x} \text{ as required}$$

$$= \text{RHS}$$

c) i) $x = 20t \cos \alpha$

$y = -5t^2 + 20t \sin \alpha$

$x = 20, \quad 20 = 20t \cos \alpha$

$t = \frac{1}{\cos \alpha}$ ✓

$\therefore h = -5 \cdot \frac{1}{\cos^2 \alpha} + 20 \cdot \frac{\sin \alpha}{\cos \alpha}$

$h = -5 \sec^2 \alpha + 20 \tan \alpha$ ✓

ii) $h = -5(\tan^2 \alpha + 1) + 20 \tan \alpha$

$h = -5 \tan^2 \alpha + 20 \tan \alpha - 5$ ✓

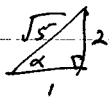
let $\tan \alpha = u$

$h = -5u^2 + 20u - 5$

max is at axis of symm: $u = \frac{-b}{2a} = \frac{-20}{-10}$

max h is at $\tan \alpha = 2$ ✓

iii) $\tan \alpha = 2$



$\sin \alpha = \frac{2}{\sqrt{5}}$

$\cos \alpha = \frac{1}{\sqrt{5}}$

$x = 20, \quad t = \frac{1}{\cos \alpha} = \sqrt{5}$

$\therefore x = 20 \cos \alpha$
 $= \frac{20}{\sqrt{5}}$ ✓

$y = -10t + 20 \sin \alpha$

$= -10\sqrt{5} + \frac{40}{\sqrt{5}} = -10\sqrt{5} + 8\sqrt{5} = -2\sqrt{5}$

$\text{speed}^2 = \dot{x}^2 + \dot{y}^2$

$= \left(\frac{20}{\sqrt{5}}\right)^2 + (-2\sqrt{5})^2$

$= 80 + 20$

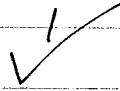
$= 100$

speed = 10 m/s ✓

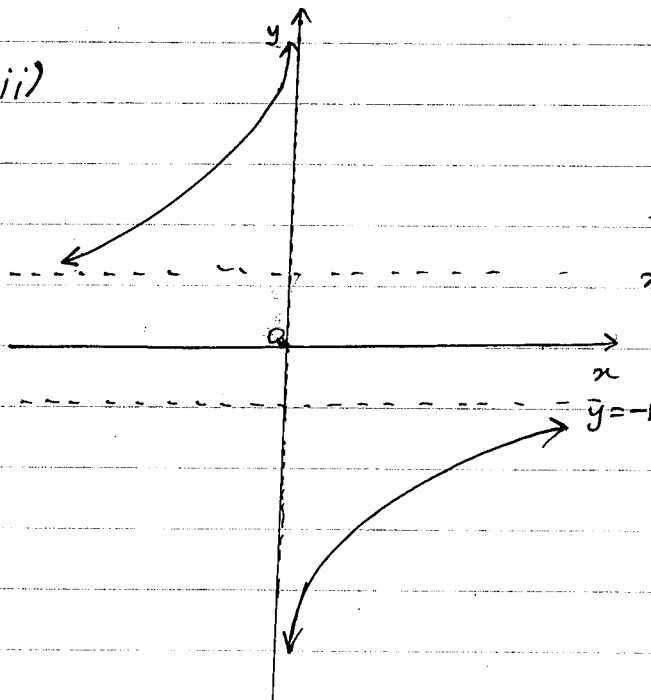
d) i) $1 - e^x = 0$

$e^x = 1$

$x = 0$



ii)



$x = 0.1, y \rightarrow -\infty$

$x = -0.1, y \rightarrow +\infty$

$x \rightarrow +\infty, y \rightarrow -1$

$x \rightarrow -\infty, y \rightarrow 1$

vertical asymptote ✓
horizontal asymptotes ✓